

Question -1

The image has 120 columns and 100 rows.

Pixel Position: The pixel with intensity "255" is located at position [52,38] (row 52, column 38).

Index = (Row Index × Number of Columns) + Column Index

Index = (52 × 120) + 38 = 6240 + 38 = 6278

The pixel with intensity "255" is at index 6278 in the 1D vector.

Question-2

I(1,0): Refers to the pixel at row 0, column 1. The value is 2.

I(3,1): Refers to the pixel at row 1, column 3. The value is 9.

I(2,2): Refers to the pixel at row 2, column 2. The value is 13.

Image Type: It is a grayscale image since the pixel values are single numbers ranging from 1 to 20.

The pixel values in this image range from 1 to 20.

Compared to grayscale range (0 , 255), these values are very low.

Therefore, this is dark image since all pixel values fall in the lower end.

Question -3

```
from matplotlib.image import imread

import matplotlib.pyplot as plt

# Function to flip and flop the image

def flipflopim(data_dir):

    # Read the image from drive

    original_image = imread(f"{data_dir}")

    # Get the
    height and width of the image
```

```
height = len(original_image)

width = len(original_image[0])

# axis for plotting the images

fig, axs = plt.subplots(nrows=1,ncols=3, figsize=(10,5))

# display
the original image

axs[0].imshow(original_image)

axs[0].set_title('Original Image');

axs[0].axis('off')

# Allocate memory for the flip image

flipped_image = original_image.copy()

# Iterate
through each pixel of the original image

for i in range(height):

    for j in range(width):

# set
corresponding values to the flip image from the original image

        flipped_image[height-1-i][j] = original_image[i][j]

# display
the flipped image

axs[1].imshow(flipped_image);

axs[1].set_title('Flipped Image');

axs[1].axis('off')

# Allocate memory for the flipflop image

flipflopped_image = original_image.copy()

# Iterate
through each pixel of the flipped image

for i in range(height):
```

```
        for j in range(width):  
            # set  
            corresponding values to the flipflop image from the flip image  
            flipflopped_image[i][width-1-j] = flipped_image[i][j]  
  
# display  
the flipflopped image  
  
axs[2].imshow(flipflopped_image);  
  
axs[2].set_title('FlipFlopped Image');  
  
axs[2].axis('off')  
  
return flipflopped_image
```

```
# Mount google drive  
  
from google.colab import drive  
  
drive.mount('/content/drive',force_remount=True)  
  
from matplotlib.image import imread  
  
import matplotlib.pyplot as plt  
  
data_dir = '/content/drive/My Drive/ComputerVision/grayscale.jpg'  
  
# calling the flipflopim function to flip and flop the  
original image  
  
flipflopim(data_dir)
```